

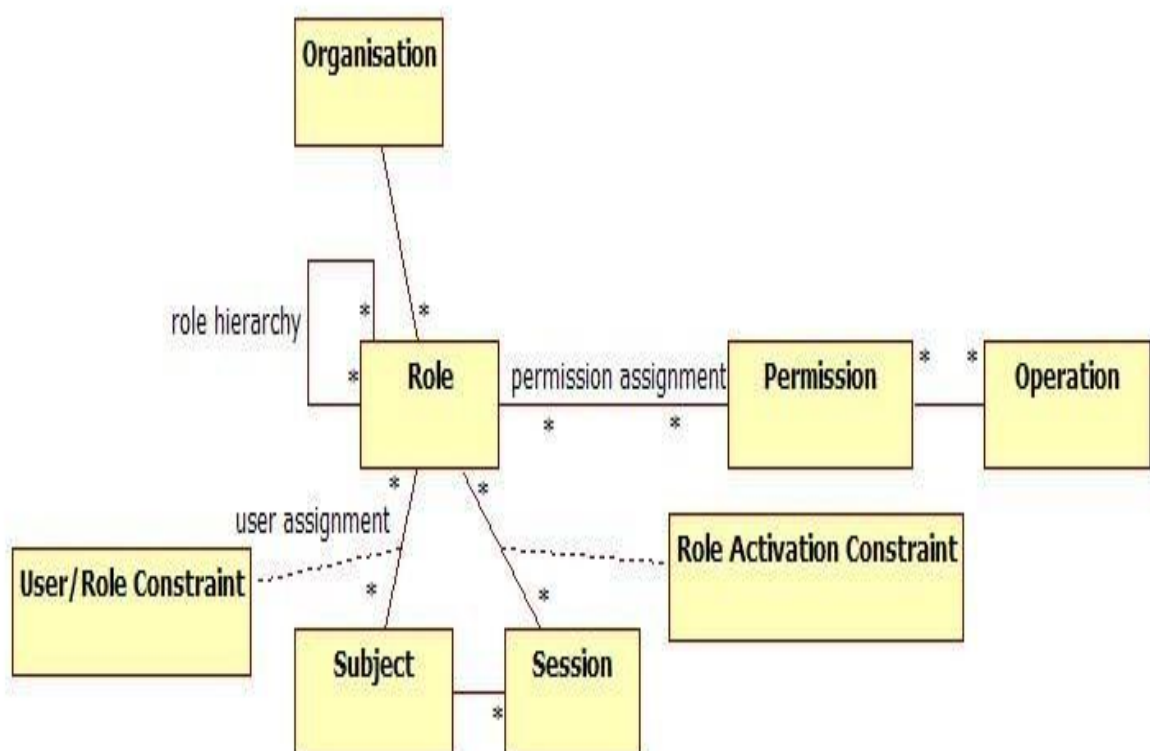
Project Design Phase Solution Architecture

Date	
Team ID	
Project Name	Optimizing User, Group, and Role Management with Access Control and Workflows
Maximum Marks	4 Marks

Solution Architecture

The proposed solution is developed on the ServiceNow platform to simplify user, group, and role management while ensuring secure access to project resources. The architecture integrates user management, access control, workflow automation, and data storage into a single cloud-based platform. Administrators manage users, groups, and roles through the ServiceNow interface, while Access Control Lists (ACLs) enforce security. Flow Designer automates approvals, notifications, and task assignments, and all project information is stored securely in ServiceNow tables.

The architecture provides a centralized environment that improves security, reduces manual effort, and enables efficient project management.



Architecture Components

User Interface

The ServiceNow Web Interface serves as the primary interaction point for administrators, project managers, and team members. It provides forms, dashboards, lists, and navigation menus that allow users to manage records efficiently.

Application Layer

The application layer contains the core business functionality of the system. It includes modules for user management, group management, role assignment, Access Control Lists (ACLs), Business Rules, and Flow Designer. These components process user requests and enforce organizational policies.

Security Layer

Security is implemented using Role-Based Access Control (RBAC) and Access Control Lists (ACLs). These mechanisms ensure that users can access only the records and fields permitted by their assigned roles. Authentication and authorization help protect sensitive project information from unauthorized access.

Database Layer

All user information, groups, roles, project records, and task records are stored in ServiceNow tables. The centralized database enables efficient data retrieval, record management, and reporting while maintaining data integrity.

Workflow Automation

Flow Designer automates repetitive administrative tasks such as project approvals, task assignments, and notification emails. Automation reduces manual work, improves consistency, and ensures timely communication among users.

Notification Service

The notification service informs users about approvals, task assignments, record updates, and workflow completion through automated email notifications generated by the platform.

References

1. <https://docs.servicenow.com/>
2. <https://developer.servicenow.com/>
3. <https://www.servicenow.com/docs/>